MILITARY SPECIFICATION SHEET
BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5112/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the lithium sulfur dioxide non-rechargeable battery type described herein shall consist of this specification and the latest issue of MIL-B-49430(ER).

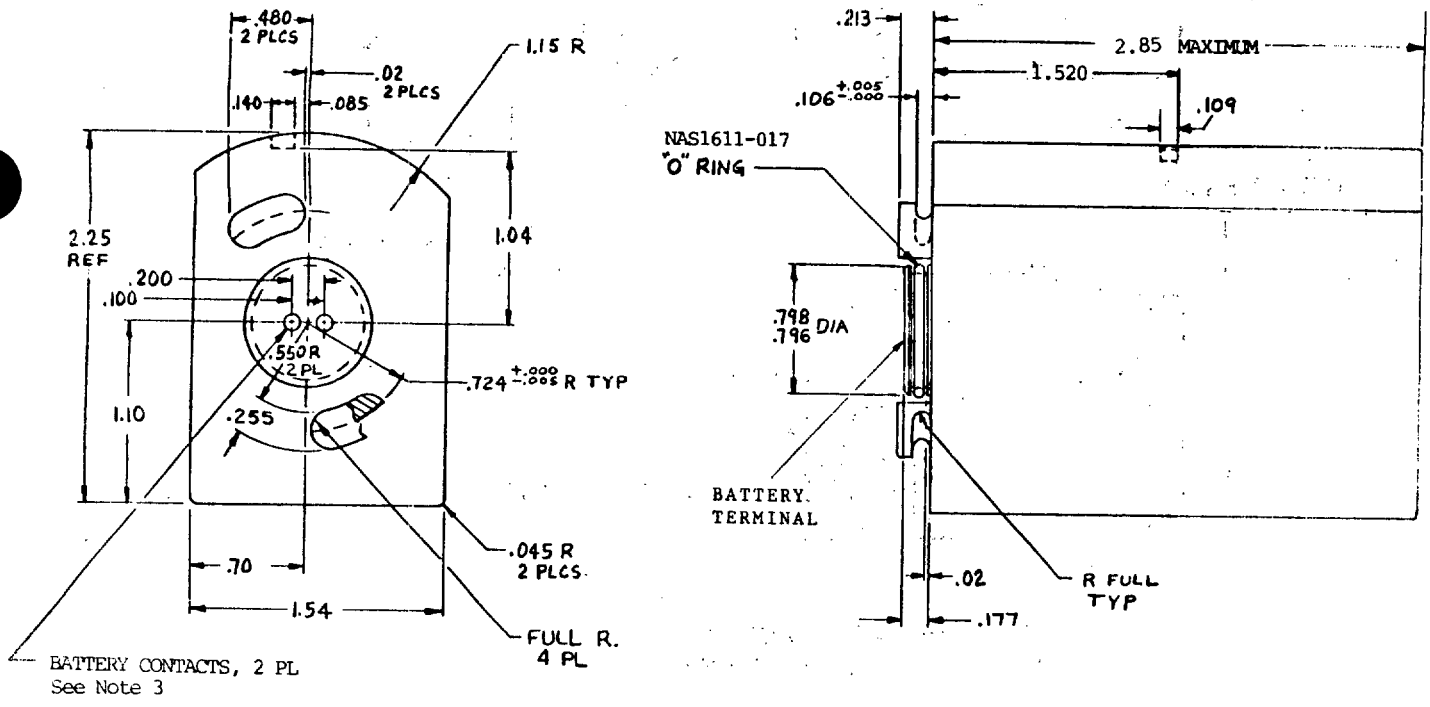


FIGURE 1

AMSC N/A

FSC 6135

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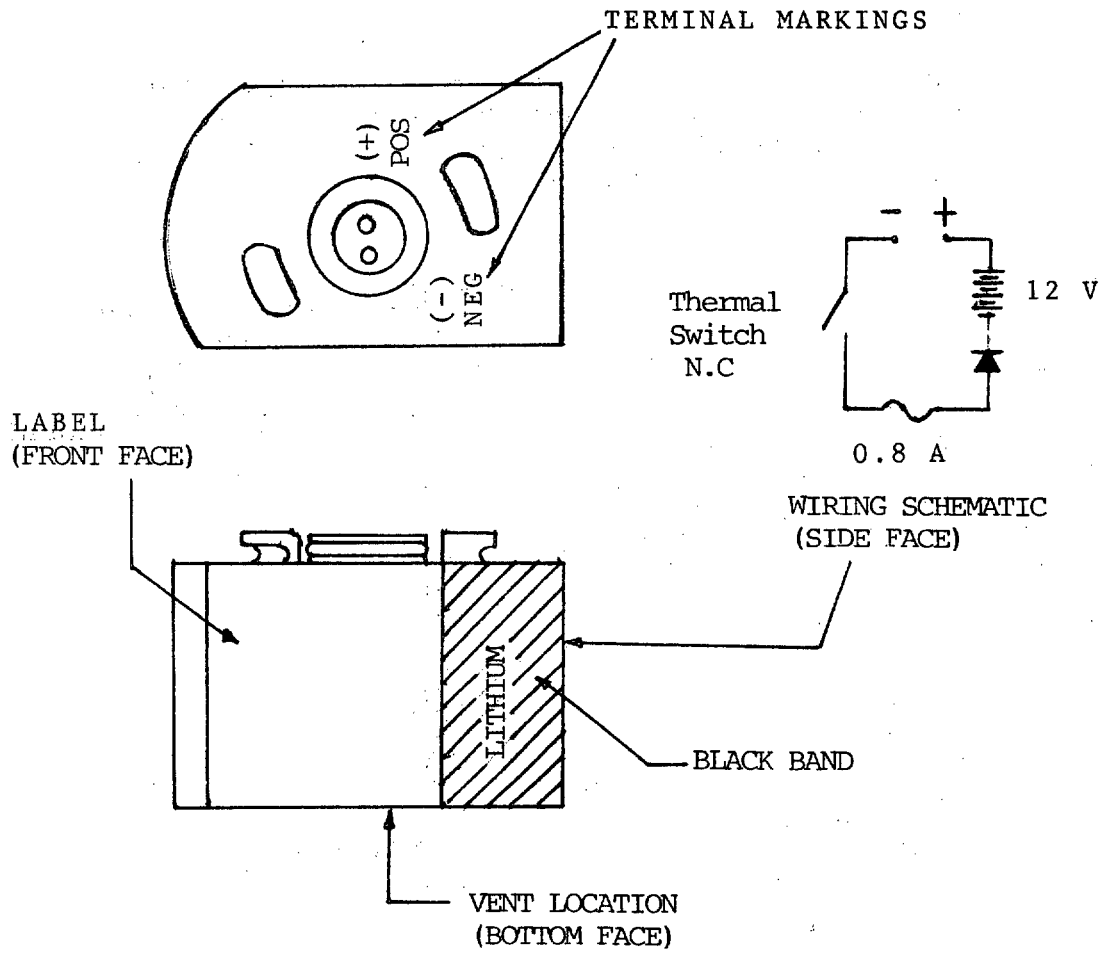


FIGURE 2

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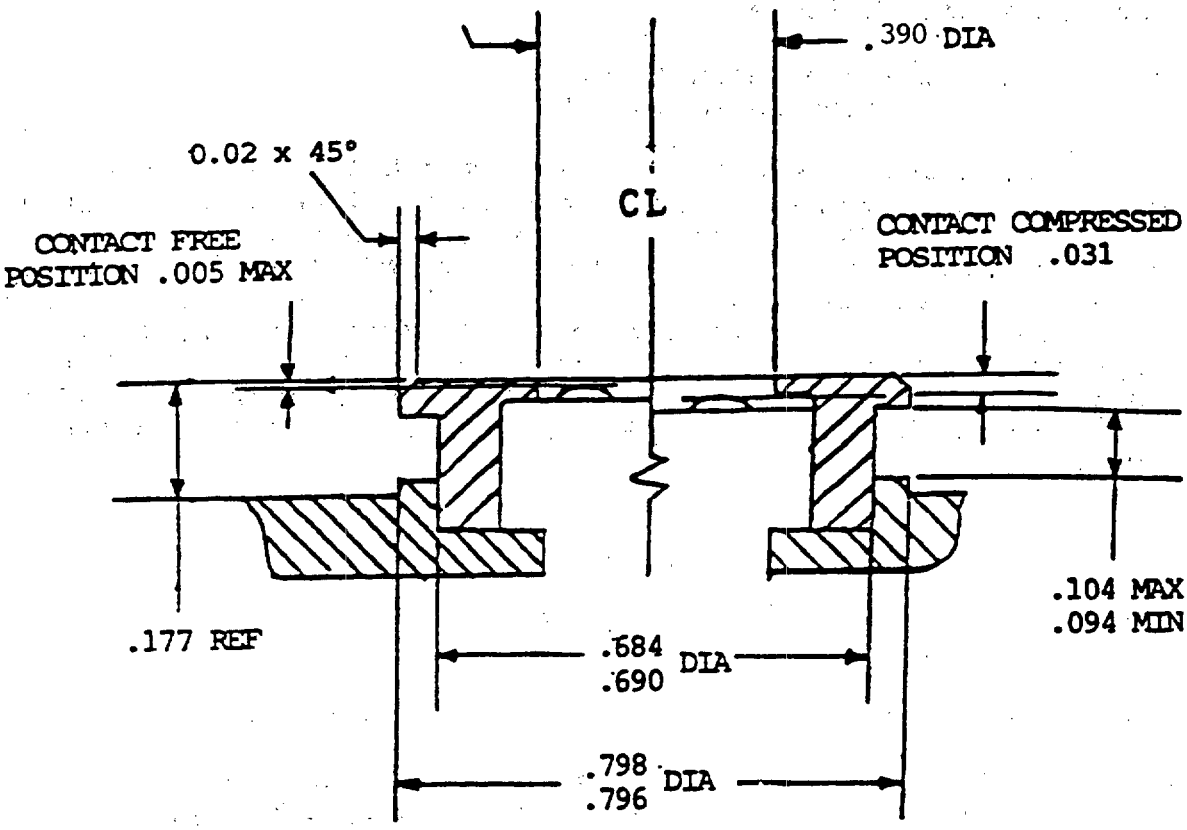


FIGURE 3

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DRAWING NOTES:

1. All dimensions are in inches.

2. Tolerances are: 2 Place Decimals ±.01
3 Place Decimals ±.005

3. Battery contacts shall be constructed from tubular, oval head rivets conforming to MS16535-11. The contacts shall be nickel plated in accordance with QQ-N-290, Class 1, Grade C. Then the contacts shall be gold plated in accordance with MIL-G-45204, Type II, Grade C, Class 2 approximately .00144 grams, .0456 square inches. The contacts shall be mounted on a spring loaded surface. The contacts shall require a minimum mating force of 15 ounces per contact. The contacts shall not protrude above the surface of the battery terminal. The contacts diameter shall be .109±.005 inch and the contact surface shall be smooth.

REQUIREMENTS:

Dimensions and configuration: See Figure 1 and Figure 3.

Electrical Fuse: The battery shall contain an 0.80 ampere time delay fuse in series with the cells in the negative leg.

Diode: The battery shall contain a blocking diode wired in series with the cells.

High Temperature Switch: A high temperature switch or thermal fuse shall be employed internal to the battery.

Maximum voltage: 12.2 volts

Minimum voltage (for capacity): 8.0 volts

Terminals: 2 pin contact, coupler interface, "O" ring sealed (see Figure 1 and 3).

Weight (maximum): 0.40 pounds (180 gms).

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Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity in Hours to 8.0 Volts
I	3.00
L	1.50
H	5.50
HT	5.30
LT	1.30
T	2.80
D	2.80
ILR	1000

Initial voltage delay: During the I, H, HT, T, D, and ILR tests, initial voltages below the 8.0 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests initial voltage below 8.0 volts cannot exceed a 60.0 second duration.

Venting Mechanism: The battery shall incorporate a venting mechanism that shall prevent the battery case from exploding or rupturing into fragments when cell venting occurs. The venting mechanism shall be located at the bottom face (see Figure 2.0). Venting mechanism shall not become separated from the battery case when activated by cell venting.

METHOD OF EXAMINATION AND TEST:

Capacity tests:

1. Storage: Details on storage conditions for all specified capacity tests are described in the basic specification.

2. Discharge:

Cells: Discharge at 0.65 amps constant current at 70±5°F for a minimum of 2.7 hours to 2.00 volts cut-off. Continue discharge to zero volts, then discharge at 0.65 amperes constant current for 2.7 hours into voltage reversal.

Batteries: For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at 0.65 amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through 0.65 amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I test the batteries shall be force discharge at 0.65 amperes constant current for 3.0 hours immediately after zero volts is reached.

There shall be no venting, fire or explosion when the batteries are forced discharged. For the ILR test a fixed resistane of 5600 ohms shall be used to a cut-off voltage of 8.0 or 1000 hours.

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Closed circuit voltage: Load resistance of 20 ohms shall be used between battery terminals. Voltage shall be above 8.0 volts within 5 seconds.

Venting Mechanism: The battery venting mechanism shall be monitored during the Safety Feature Test. During cell venting the battery venting mechanism shall release the pressure build up in the battery case and shall meet the performance requirements specified herein.

CUSTODIAN:

ARMY-ER

PREPARING ACTIVITY:

ARMY-ER

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